

Minor Project Synopsis Report
Measuring Pedagogical Alignment In Learning Project

Category: University Based Project

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ABSTRACT

This research focuses on evaluating the pedagogical alignment of mobile learning applications used by preschool and primary school children. With the rapid increase in educational apps, it has become important to examine whether these apps truly support learning or just provide entertainment through animations and rewards.

The study uses established educational frameworks such as Bloom's Taxonomy and evidence-based learning principles to analyze the effectiveness of these apps. A structured evaluation framework will be developed to measure how well the apps support cognitive development, engagement, and meaningful learning outcomes.

The research aims to identify high-quality educational apps and highlight the limitations of poorly designed ones. The findings will help educators, parents, and developers improve the quality of digital learning tools.

1. INTRODUCTION

In today's digital world, mobile learning applications have become an important part of early education. Children use these apps to learn basic skills such as reading, writing, counting, and problem-solving. However, not all learning apps are equally effective.

Pedagogical alignment refers to how well learning objectives, teaching methods, and assessments are connected with each other. A well-designed app should follow proper learning theories and provide meaningful educational experiences.

This research studies whether modern learning apps follow educational principles or mainly focus on entertainment. It highlights the importance of aligning technology with pedagogy to improve learning outcomes.

2. MOTIVATION

The motivation behind this research comes from the increasing dependence on mobile learning apps by children, parents, and teachers.

Many apps appear attractive and engaging but lack proper educational structure. There is no standard method to evaluate whether these apps actually help in learning. This creates confusion for users when selecting the right app.

Therefore, this research is motivated by the need to

- Ensure quality education through digital platform
- Provide a reliable evaluation system
- Help users choose effective learning apps

3. LITERATURE REVIEW

Previous research shows that integrating educational theories into digital learning is essential for effective learning.

- Bloom's Taxonomy explains different levels of learning:

- Remembering
- Understanding
- Applying
- Analyzing
- Evaluating
- Creating
- Evidence-based learning principles highlight:
- Active participation
- Feedback mechanisms
- Practice and repetition
- Learner engagement

Studies reveal that many learning apps only focus on basic memorization and do not support higher-level thinking skills. Some apps include interactive features but fail to align with proper learning objectives.

Recent studies also emphasize the importance of adaptive learning and personalization for better results

4. GAP ANALYSIS

Even though many learning apps are available, several issues still exist:

- No standard framework to measure educational quality
- Apps focus more on entertainment than learning
- Lack of higher-order thinking skills
- Limited feedback and interaction
- Poor personalization features

This research aims to fill these gaps by creating a structured evaluation system based on educational theories.

5. PROBLEM STATEMENT

Many mobile learning applications claim to be educational, but there is no proper system to verify their effectiveness. Most apps attract users through visuals and rewards but fail to provide meaningful learning experiences.

The main problem is the absence of a reliable method to measure pedagogical alignment. This research aims to solve this issue by developing a systematic evaluation framework.

6. OBJECTIVES

The main objectives of this research are:

- To understand pedagogical alignment and learning theories
- To study the structure of learning applications
- To apply Bloom's Taxonomy for evaluation

- To develop a framework for assessing apps
- To compare different learning apps
- To suggest improvements for better app design

7. Tools/Technologies Used

The following tools will be used in this research:

- Mobile Apps:
 - Duolingo
 - Khan Academy Kids
- Software Tools:
 - Microsoft Excel / Google Sheets (data analysis)
 - MS Word / Google Docs (documentation)
- Other Resources:
 - Research papers and journals
 - Educational websites

8. METHODOLOGY

The research will follow a systematic process:

1. Selection of Apps – Choose 5–10 educational apps
2. Framework Design – Create evaluation criteria
3. Data Collection – Analyze features like content and interaction
4. Evaluation – Assign scores based on learning principles
5. Analysis – Compare results
6. Validation – Verify results with research and experts
7. Conclusion – Provide final findings and suggestion

10. Expected Outcomes

The research is expected to produce:

- A clear evaluation framework
- Identification of best and worst learning apps
- Suggestions for improving app design
- Better understanding of digital learning quality

11. References

- Research papers on Bloom's Taxonomy
- Journals on educational technology
- Articles on evidence-based learning
- Official websites of learning apps
- Academic books on pedagogy